DATA SCIENCE II | PROJECT

# Monitoring Deforestation and Land-Use Change in the Amazon

## Using Deep Learning on Satellite Imagery

OluwaKoretimi Adebowale | Josselin Jimenez | Mehrab U Azan

Multi-label tile classification · NDVI feature engineering · Transfer learning · Grad-CAM

*Background: illustrative synthetic imagery, not real satellite data*

# Progress is real, but the forest is still shrinking

**5,796 km²**

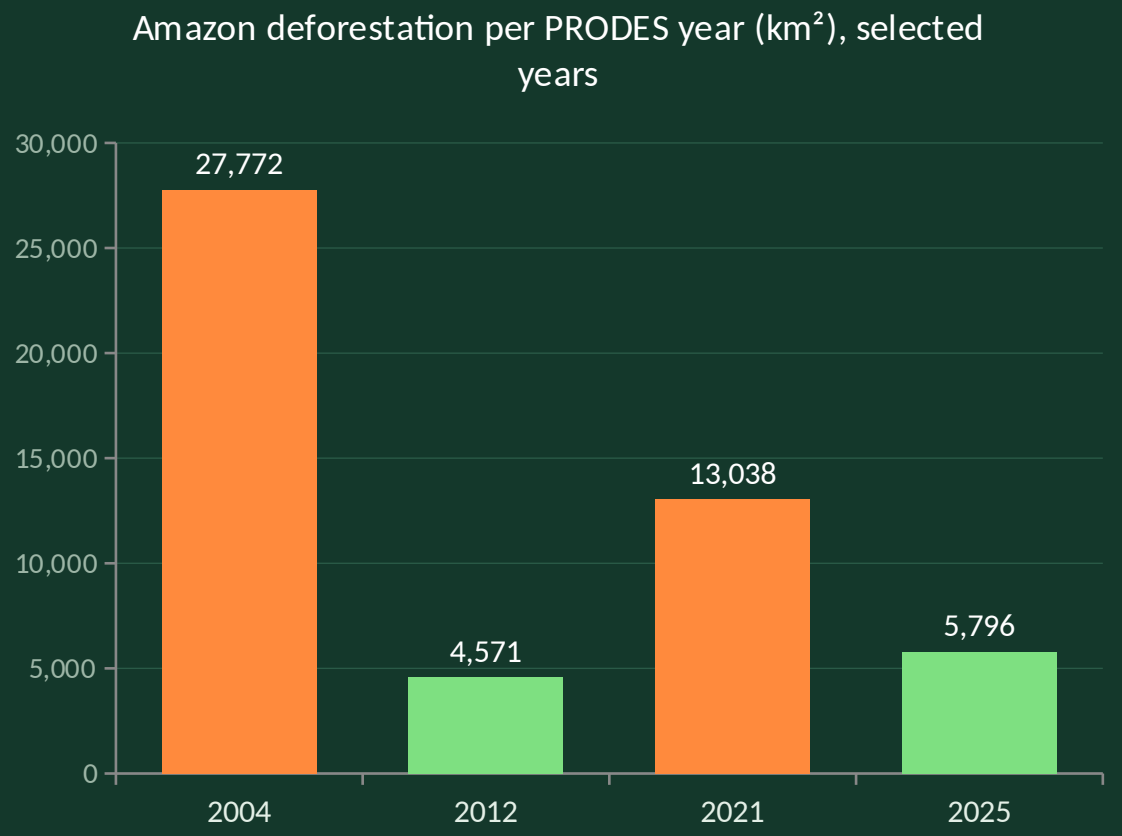
PRODES deforestation in 2025, down 11.08% from 2024

**2,874 km²**

DETER alerts, Aug 2025 to Jul 2026: lowest since 2015 (36% below prior year)

**Every year**

Enforcement still depends on spotting new clearing early




*Sources: INPE PRODES / DETER via O Liberal (2025) and Mongabay (Aug 2026); 2004, 2012, 2021 from INPE PRODES.*

# Why we need machines to watch the forest

## Scale

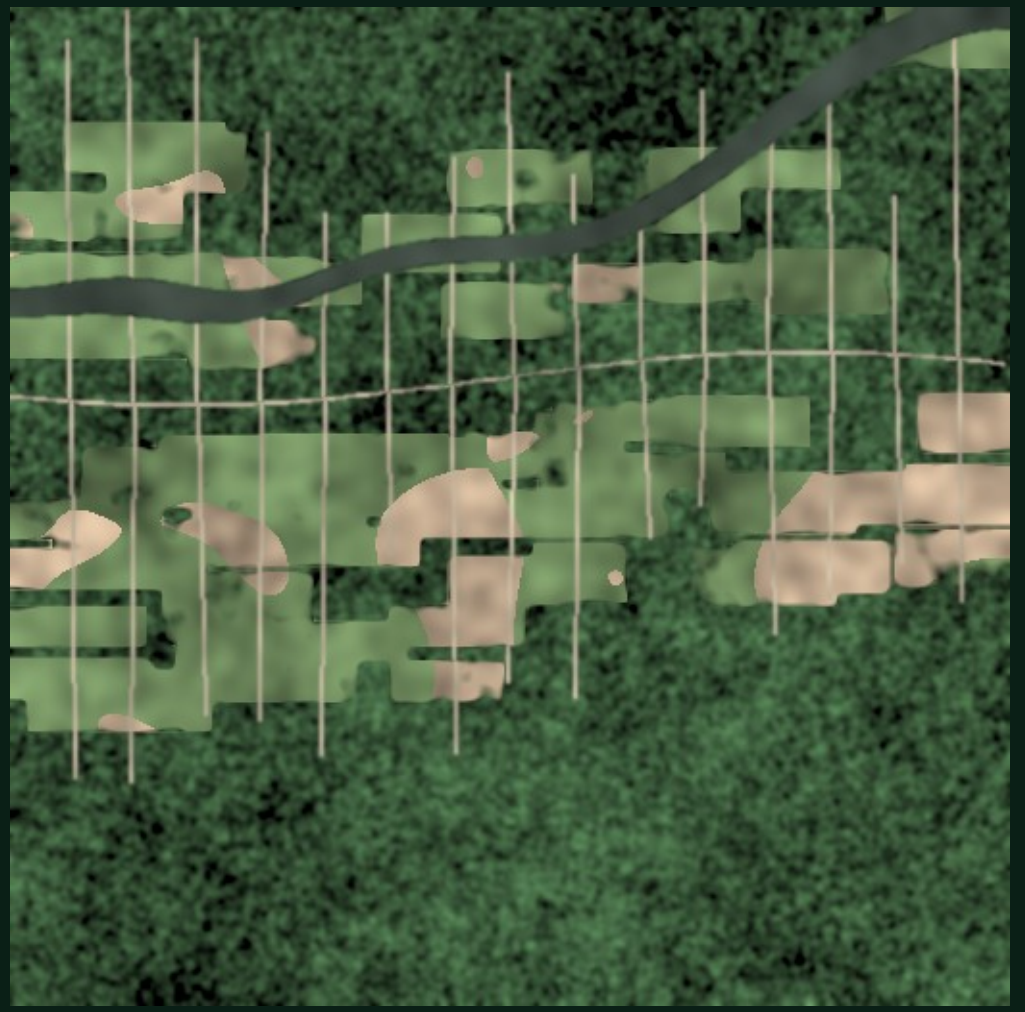
The basin is far too large for analysts to review tile by tile.

## Speed

Automated tagging can surface new clearing in near real time.

## Consistency

One model applies the same rules to every tile, every pass.



*Illustrative synthetic tile: clearings branching off a road and river*

# The data: Planet's Amazon-from-space image chips

**~40,000**

labeled 256×256 px chips

**4 bands**

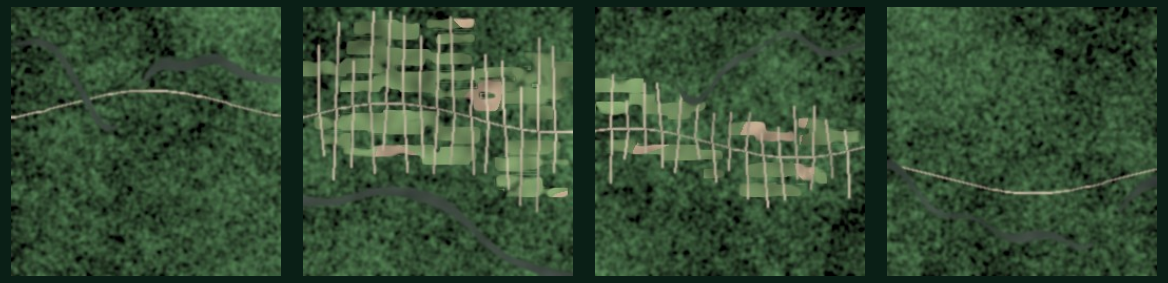
Red, Green, Blue, Near-Infrared

**17 labels**

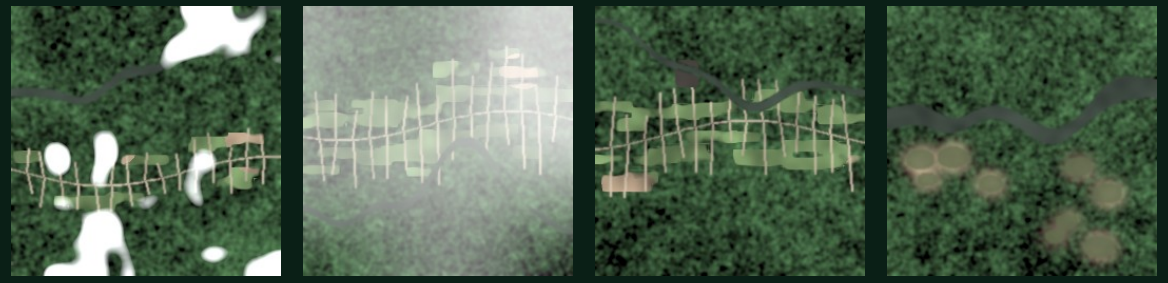
atmosphere + land cover + land use

**~20 GB**

Kaggle: Understanding the Amazon from Space



primary forest | agriculture | road | water

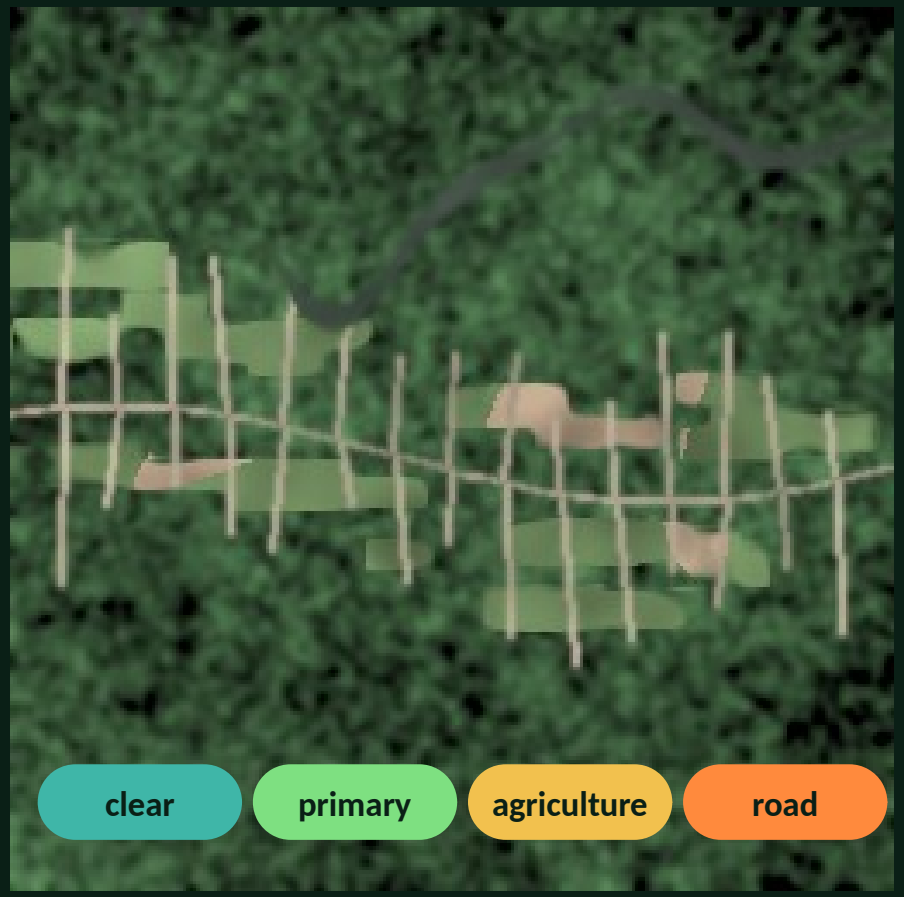


partly cloudy | haze | slash & burn | artisanal mine

Each tile can carry several labels at once, e.g. clear + primary + agriculture + road.

*Tiles shown are illustrative synthetic renderings of each label type, not images from the dataset.*

# A multi-label problem, not a single-class one




*One tile, four tags (illustrative)*

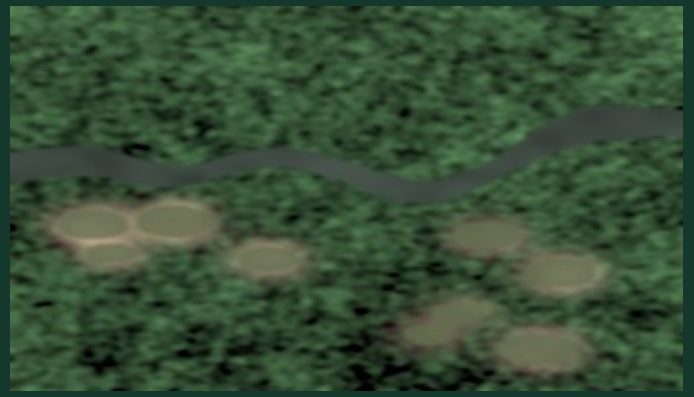
## Single-label

- Softmax output
- Classes compete: probabilities sum to 1
- Forces a choice: forest OR farm

## Multi-label (ours)

- 17 independent sigmoid outputs
- Binary cross-entropy per label
- Forest AND road AND haze can all be true

# Three things that make this hard



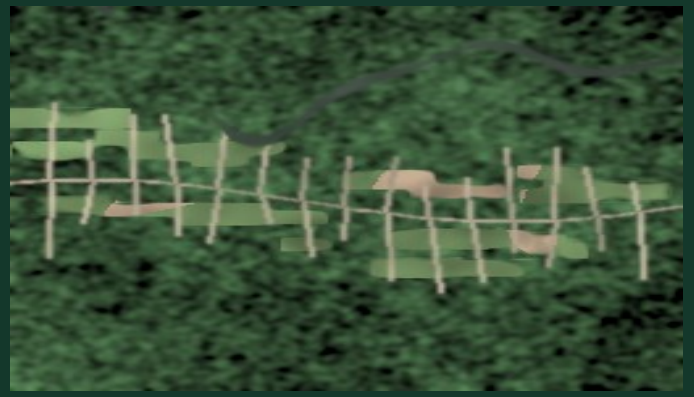
## Severe class imbalance

Rare, high-impact labels such as blow_down and artisanal_mine appear in under 1% of tiles, while primary forest dominates.



## Clouds and haze

Atmospheric conditions hide the ground and can mimic bare soil, so the model must learn to see through or around them.



## Mixed land use in one tile

A single 256×256 chip often holds forest, road and farmland together, so boundaries are fuzzy.

# Our approach: from raw chips to explained predictions

### Satellite chips

4-band R,G,B,NIR tiles, 256×256

›

### Preprocess + NDVI

Normalize bands; add vegetation index channel

›

### Augment + rebalance

Rotate, flip, oversample rare labels

›

### CNN classifier

Sigmoid outputs, BCE loss per label

›

### Tune thresholds

Per-label cutoffs optimised for F2

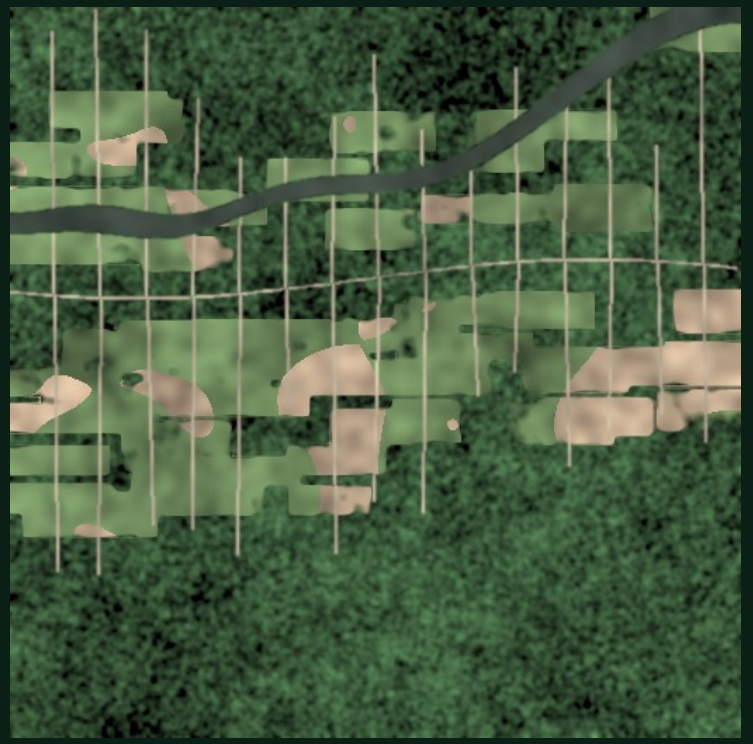
›

### Evaluate + explain

F2, recall, Grad-CAM error analysis

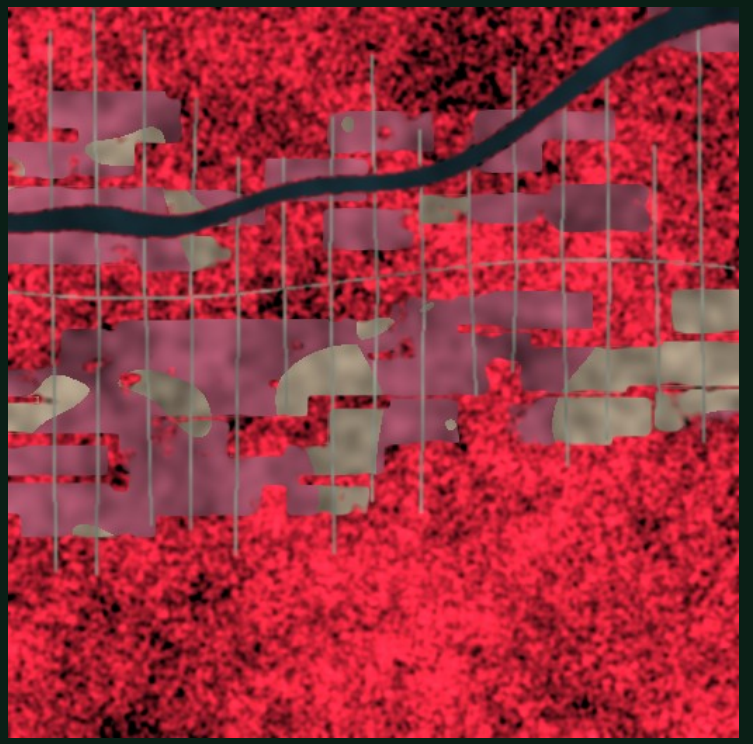
Everything is trained and evaluated with the same split so improvements can be attributed to one change at a time.

# NDVI makes forest loss visible



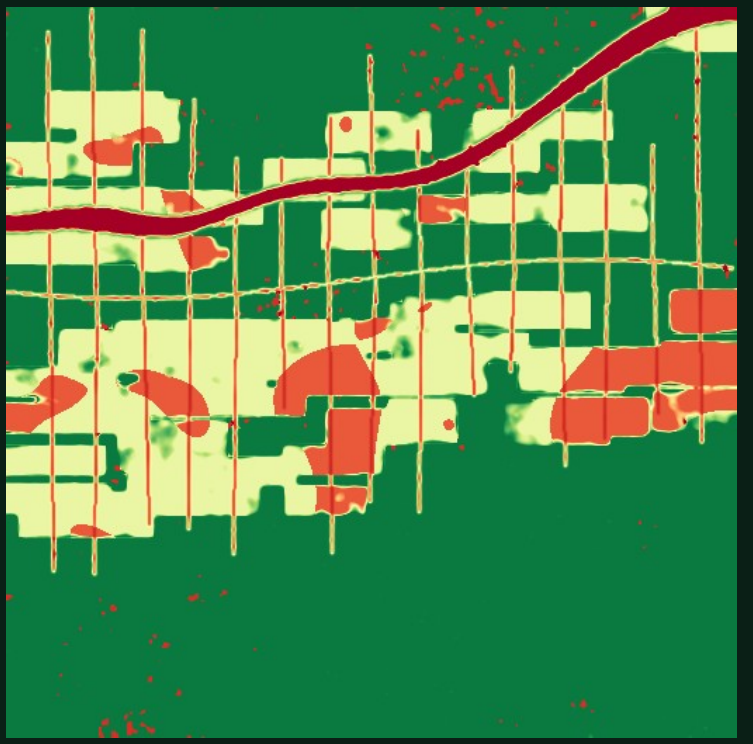
**Visible (RGB)**

What a person sees



**Near-infrared false colour**

Healthy canopy glows red



**NDVI channel**

Green = dense canopy, red = cleared

**NDVI = (NIR – Red) / (NIR + Red)** dense canopy near +1, pasture and soil near 0, water below 0

# Fighting imbalance: augment, oversample, reweight

## Rotations and flips

Top-down imagery has no fixed 'up', so every orientation is a valid new sample.

## Oversample rare labels

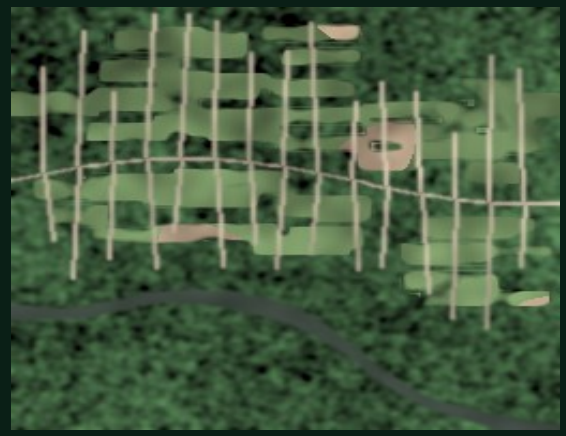
Draw tiles with mining, logging and burning more often during training.

## Class-weighted loss

Penalise missed rare labels more than missed common ones.

## Per-label thresholds

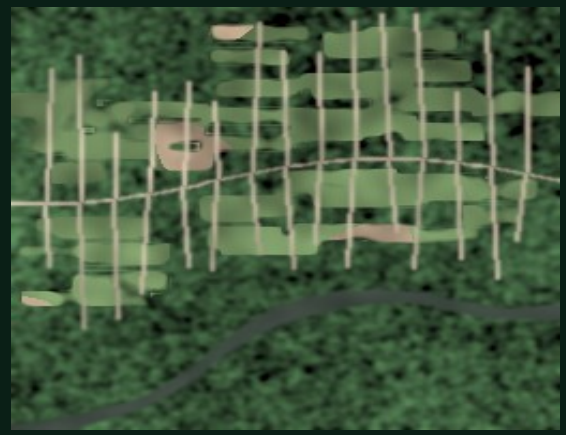
Lower the cutoff where recall matters, tuned for F2.



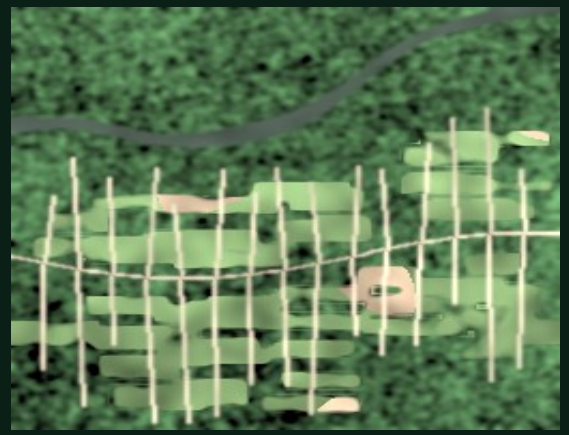
original



rotate 90°



flip left-right



flip up-down + brighten

# Models and the experiment grid

## Baseline: CNN from scratch

Stacked conv + pooling blocks, global pooling, a 17-way sigmoid head. Shows what the data alone can teach.

## Transfer learning

ImageNet-pretrained ResNet or EfficientNet. The first convolution is widened to accept 4+ channels (RGB weights copied, NIR and NDVI initialised from their mean).

## Four experiments isolate each idea

| | RGB only | RGB + NIR + NDVI |
|---|---|---|
| Scratch CNN | A | B |
| Pretrained | C | D |

B vs A and D vs C measure the effect of NDVI; C vs A and D vs B measure the effect of transfer learning.

# Why F2? Missing deforestation costs more

$$F_2 = 5 \cdot P \cdot R \,/\, (4 \cdot P + R)$$

Recall is weighted more heavily than precision, matching how enforcement works: a missed clearing is lost forest, a false alarm is a quick check.

- Per-label and macro-averaged F2
- Precision / recall curves per label
- Confusion analysis and Hamming loss
- Grad-CAM on misclassified tiles

## Reading the metrics

- Macro F2 gives rare labels the same weight as common ones
- Accuracy alone would hide the imbalance problem
- Per-label thresholds are tuned on validation data only
- We report rare-class recall separately because that is the real mission

# What success looks like

## ≥ 0.85

macro-averaged F2 across all 17 labels

## ≥ 0.60

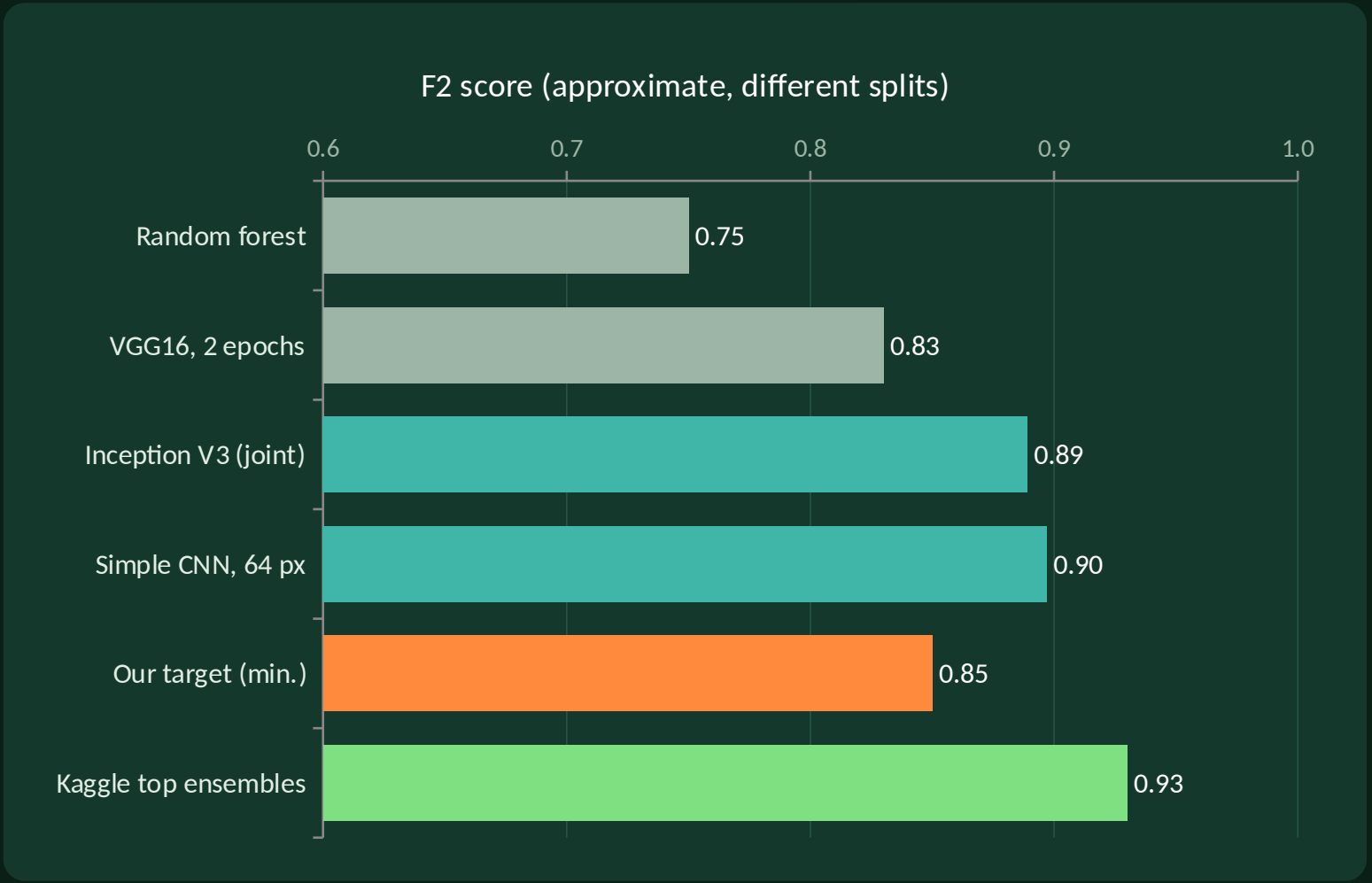
recall on mining, selective logging and slash-and-burn

## NDVI > RGB

a clear, measurable gain over the RGB-only baseline

**The recall target on rare classes matters more than the headline score.**

# Where published models land on this dataset




## Takeaways

- Even a small CNN clears our 0.85 target
- Ensembles and a soft F2 loss pushed winners past 0.93
- Our novelty is the NDVI ablation and rare-class focus, not leaderboard chasing

*Sources: GitHub mikeskaug/amazon-classifier; Stanford CS231n 2017 report; Kaggle winners' write-ups. Scores are indicative only.*

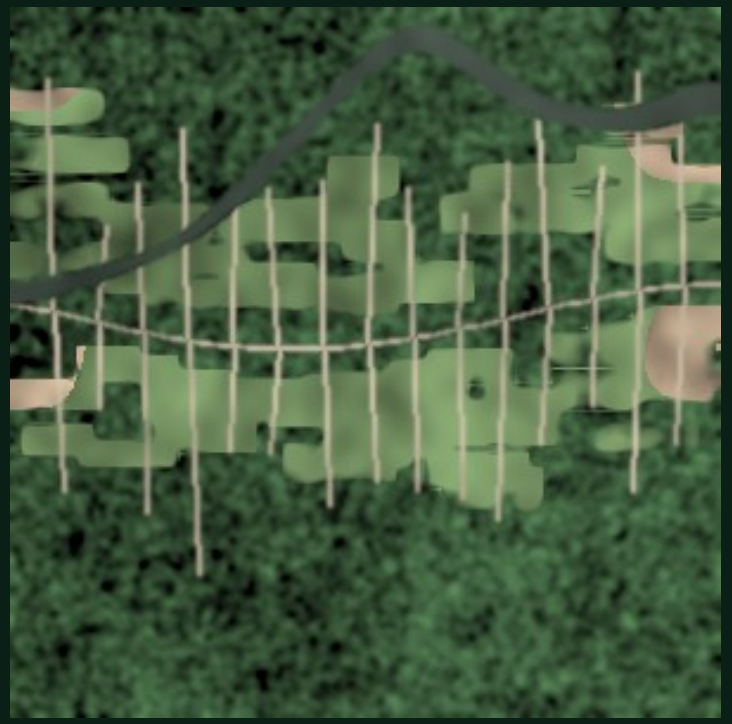
# Experiment scorecard

Fill in from your training run

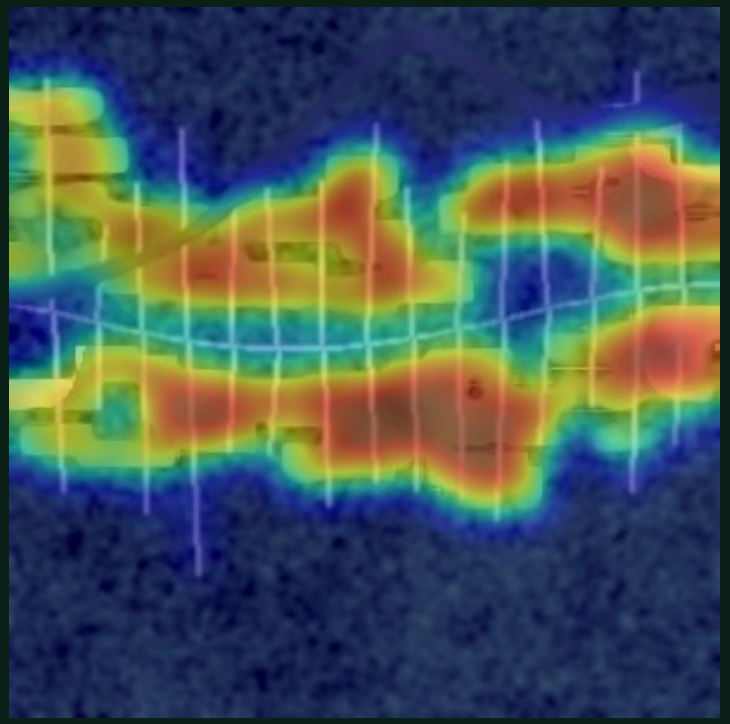
| Experiment | Macro F2 | Rare-class recall | Hamming loss |
|---|---|---|---|
| A Scratch CNN, RGB | — | — | — |
| B Scratch CNN, RGB + NIR + NDVI | — | — | — |
| C Pretrained, RGB | — | — | — |
| D Pretrained, RGB + NIR + NDVI | — | — | — |

Targets: macro F2 ≥ 0.85, rare-class recall ≥ 0.60, and D (or B) beating its RGB-only twin.

# Opening the black box with Grad-CAM



**Input tile**



**Where the model looks (illustrative)**

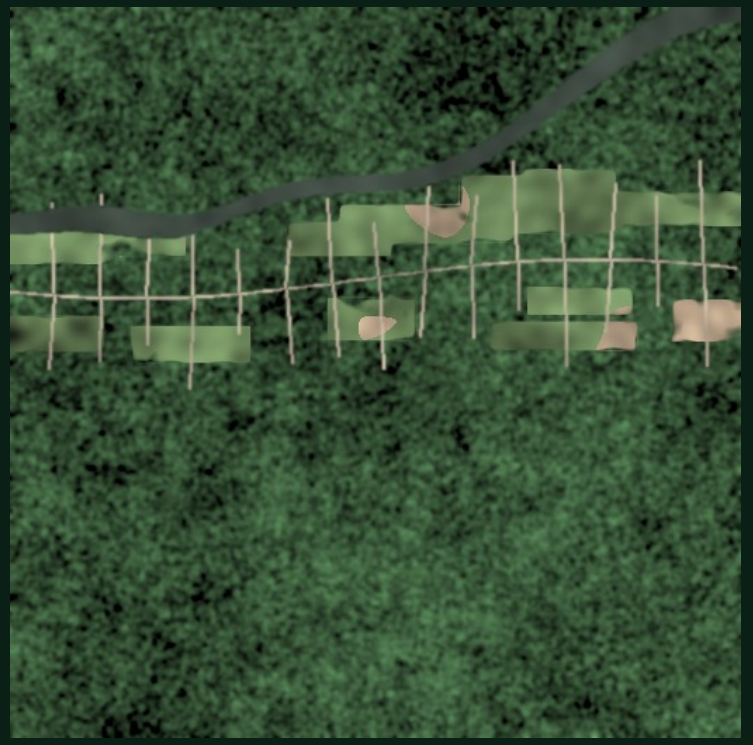
Red = strongest influence on the 'agriculture' prediction

## Error analysis questions

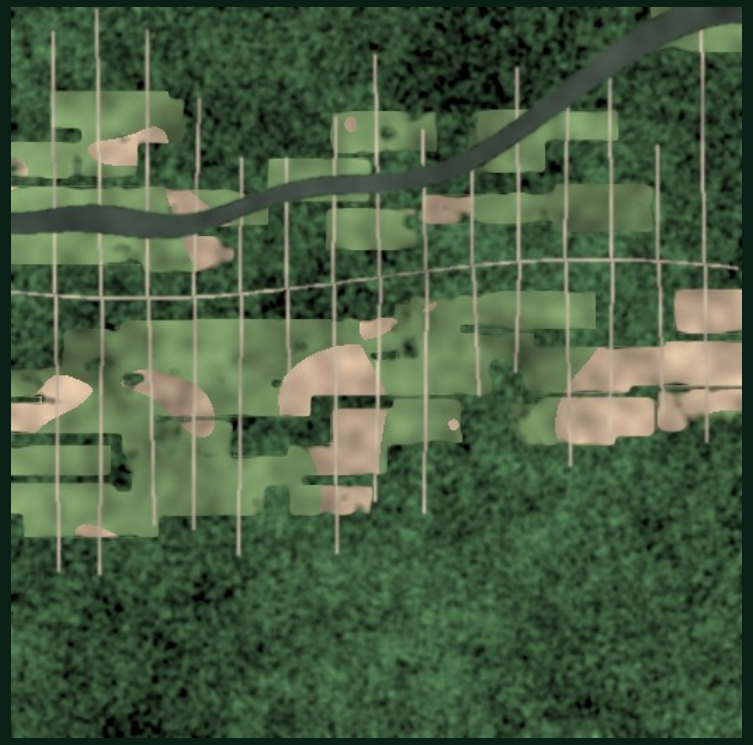
- Is the model looking at cleared land, or at cloud edges?
- Do mistakes come from mixed land use inside one tile?
- Which errors are genuinely ambiguous even to a person?

We compare RGB-only vs NDVI models on the same misclassified tiles.

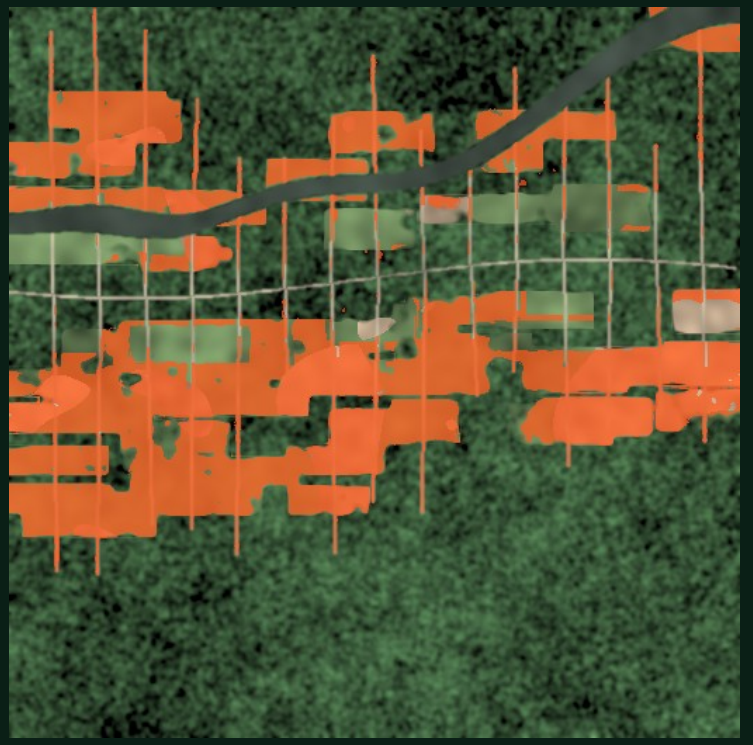
# From tile tags to land-use change maps



Earlier pass



Later pass



Flagged change

Comparing NDVI and predicted labels across two acquisitions highlights where primary forest became road, pasture or bare soil. Illustrative synthetic scenes.

# Who benefits, and how it is used responsibly



## Enforcement agencies

Rank tiles by risk so patrols and inspections go where clearing is most likely.



## Carbon accounting

Track primary-forest loss and degradation more consistently.



## Policy and NGOs

Evidence that is fast, repeatable and explainable.

Detect › Prioritise › Verify with a human › Act

Predictions are leads for human review, never automatic accusations.

# Risks we planned for

| Risk | Mitigation |
| --- | --- |
| **Cloud and haze hide the ground** | Keep haze and cloud labels; analyse cloud-driven errors with Grad-CAM |
| **Rare classes are under-learned** | Oversampling, class weights, per-label thresholds, recall-first targets |
| **Overfitting on ~40k tiles** | Augmentation, pretrained backbones, early stopping on validation F2 |
| **Results may not transfer to new regions or years** | Report limits clearly; propose temporal and spatial hold-outs as next step |

# Who is doing what

## OluwaKoretimi Adebowale

**Data and preprocessing**

Dataset setup, band normalisation, NDVI feature, label-frequency analysis, augmentation and oversampling

## Josselin Jimenez

**Modelling**

Scratch and pretrained CNNs, hyperparameters, per-label thresholds, class weights, experiment tracking

## Mehrab U Azan

**Evaluation and interpretability**

F2, precision/recall, Hamming loss, confusion matrices, Grad-CAM error analysis

Shared by all three: interpreting results, writing the report, and reviewing every section.

# Takeaways

1. A multi-label CNN can tag Amazon tiles with land use, land cover and weather in one pass.

2. An NDVI channel and rare-class focus target exactly the changes that matter to enforcement.

3. A clean 2×2 experiment shows what NDVI and transfer learning each contribute.

## Thank you. Questions?

# References and image credits


- Kaggle. Planet: Understanding the Amazon from Space. kaggle.com/competitions/planet-understanding-the-amazon-from-space
- Selvaraj, A., et al. Deep Learning Approaches for Land Cover and Land Use Classification Using Remote Sensing Imagery. Remote Sensing Applications, 2019 to 2023
- INPE. PRODES and DETER deforestation monitoring (TerraBrasilis)
- O Liberal (2025). Desmatamento na Amazônia em 2025 caiu 11,08% em relação a 2024, afirma Prodes/Inpe
- Butler, R. A. (9 Aug 2026). Amazon deforestation alerts fall to lowest level since 2013. Mongabay
- Stanford CS231n (2017). Project report on Planet Amazon classification (F2 benchmarks)
- Skaug, M. amazon-classifier (GitHub), random forest and VGG16 baselines
- *All satellite-style images in this deck are synthetic illustrations generated by the team; they are not real satellite data.*